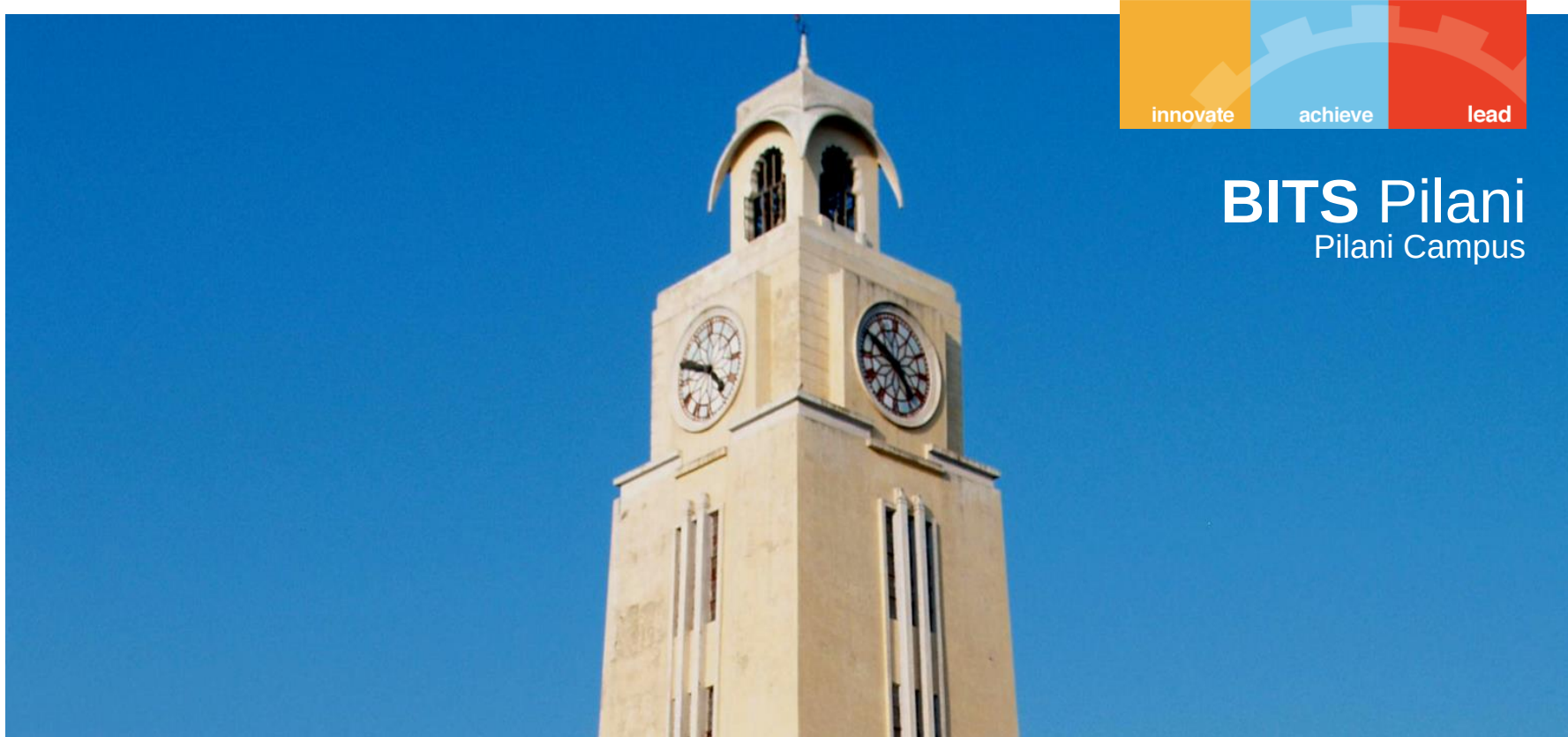




BITS Pilani
Pilani Campus



CHEM F111 : General Chemistry

Semester II: AY 2023-24

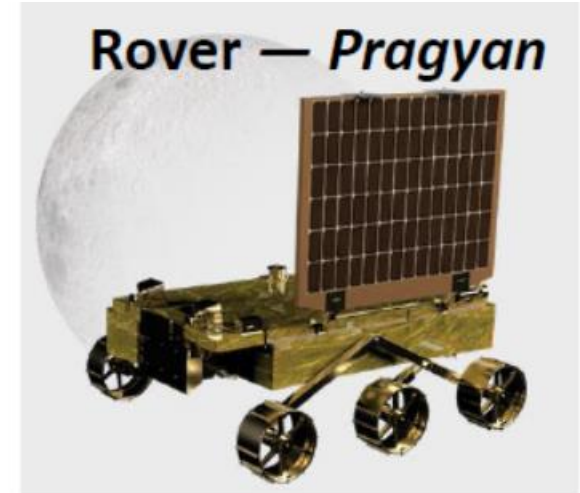
Lecture- 13, 12th February 2024, Monday

Last Class:

- Molecular orbital theory; gerade, ungerade
- Overlap integral, normalization constant, bonding and antibonding orbital energy of a diatomic molecule;
- Bond order calculation;
- Period 2 homonuclear diatoms and their molecular orbitals;
- Paramagnetism and diamagnetism

- **Spectroscopy**: Analysis of electromagnetic radiation absorbed, emitted, or scattered by atoms and molecules.
- In molecules, transitions happen between electronic /vibrational /rotational energy levels.
- Information regarding energy levels, geometry (bond lengths, bond angles), and bond strengths.
- Useful in qualitative and quantitative analysis, and in studies of kinetics, reaction flows, astronomy etc.

- ❖ Analytical, organic, inorganic use spectroscopy to:
 - i. identify new compounds
 - ii. identify intermediates
 - iii. predict reaction mechanism etc.
- ❖ Physical chemistry lab:
 - i. energy levels and transition,
 - ii. electric dipole moment, quadrupole moment, Polarizability,
 - iii. moment of inertia, NMR, fine structure etc.
- ❖ Atomic spectroscopy is the experimental technique based on which much of QM evolved



Chandrayaan 2 Large Area Soft X-ray Spectrometer:

Elemental composition of the Moon

Imaging IR Spectrometer: Mineralogy mapping and water-ice confirmation

Alpha Particle X-ray Spectrometer and Laser Induced

Breakdown Spectroscope: In-situ elemental analysis and abundance in the vicinity of landing site

The Mars Exploration Rovers were launched with the goal of searching for and analyzing rock and soils on Mars. They utilized several spectrometers to analyze samples.



Mini-TES: **mini**ature **ther**mal **em**ission spectrometer (examine rock, soil & atmosphere)

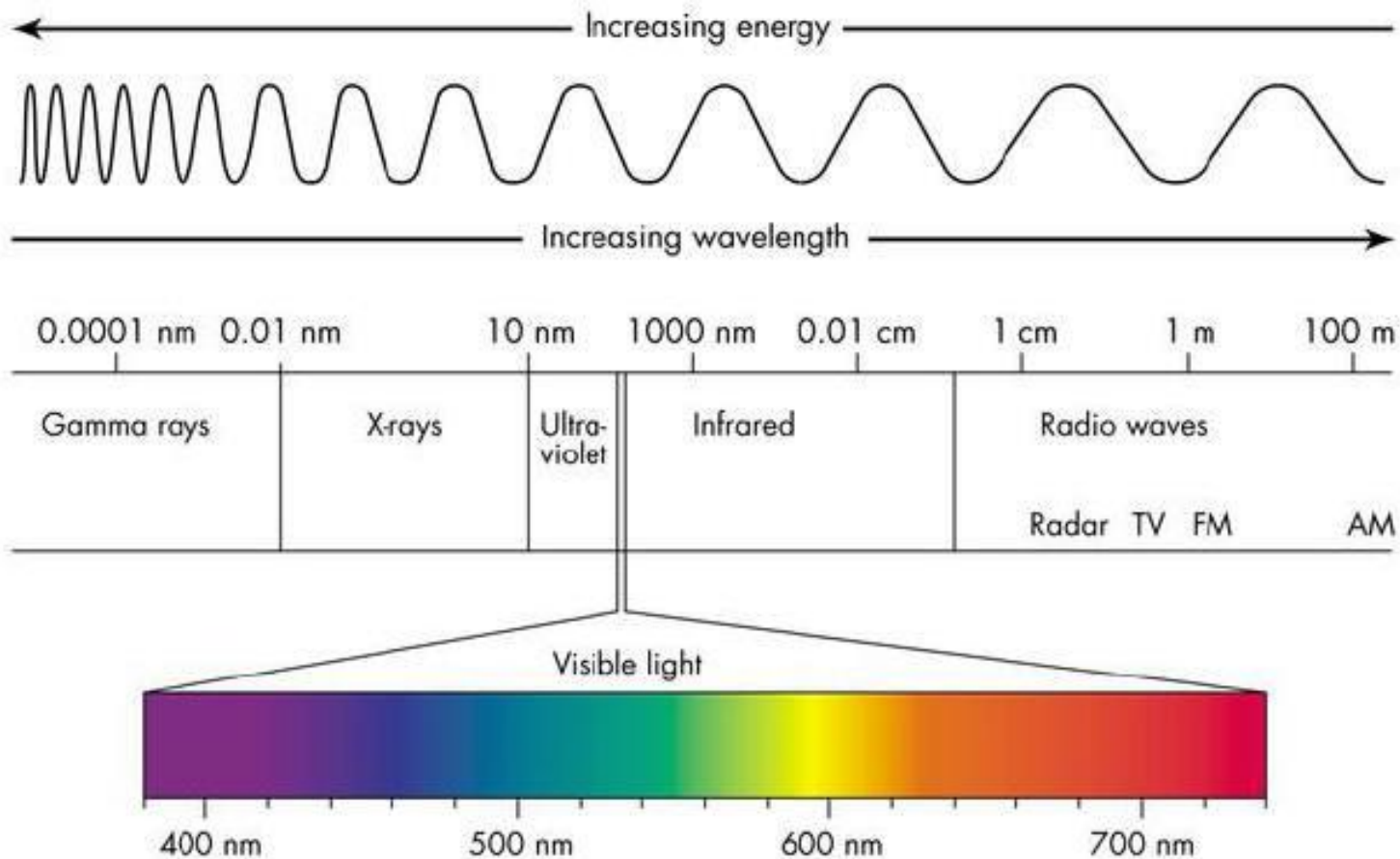
MB: **M**ossbauer **S**pectrometer (examine mineralogy of rocks & soils)

APXS: **A**lpha **P**article **X**-ray **S**pectrometer (analyze elements in rocks & soils)

Electromagnetic Spectrum



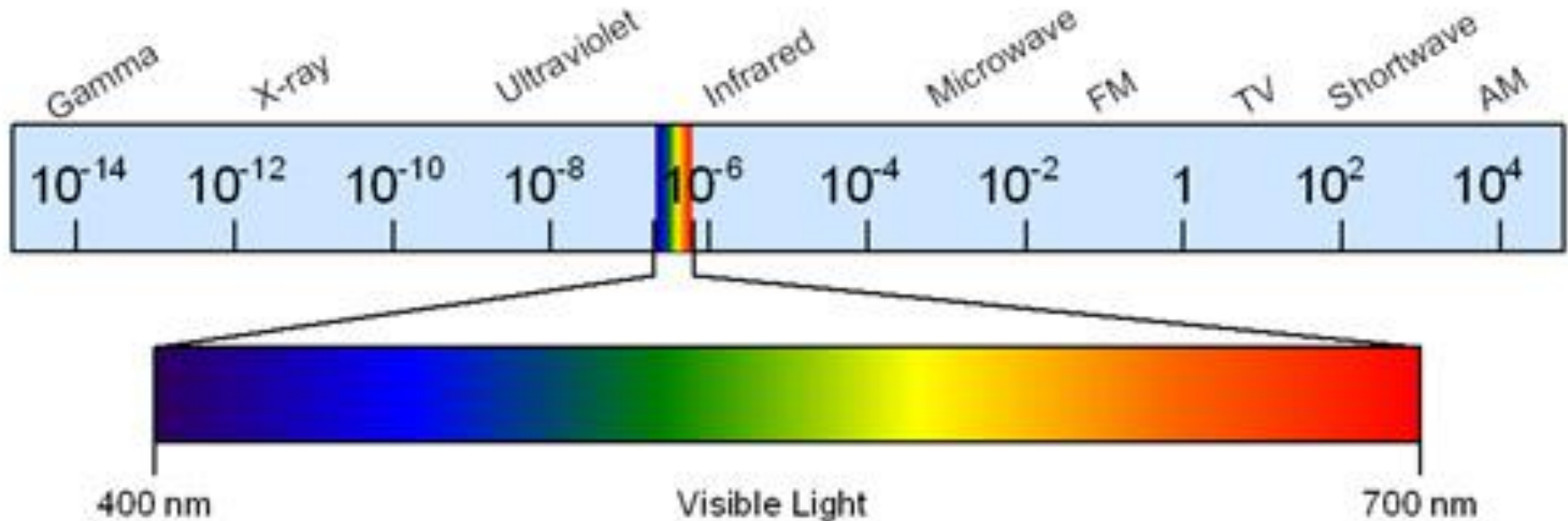
The range of energies that can be used for spectroscopy is very large and spans a large proportion of the electromagnetic spectrum.



Electromagnetic Spectrum



The Electromagnetic Spectrum (Wavelengths in Meters)



short ← ————— Wavelength (λ) ————— → long
high ← ————— Frequency (ν) ————— → low
high ← ————— Energy (E) ————— → low

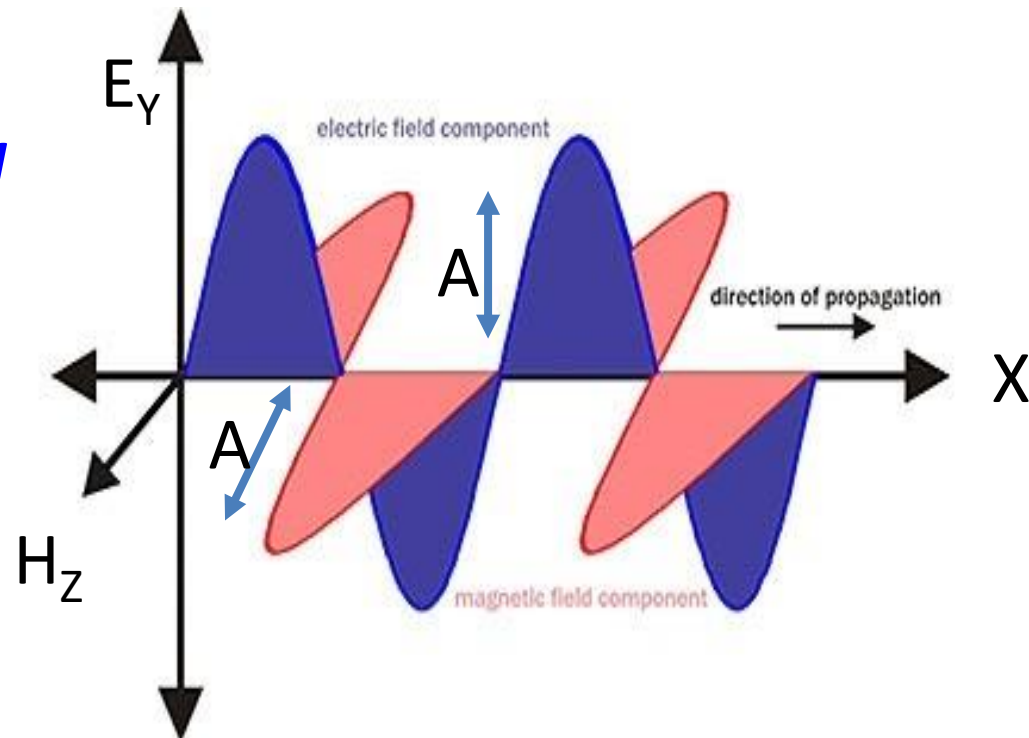
Electromagnetic radiation



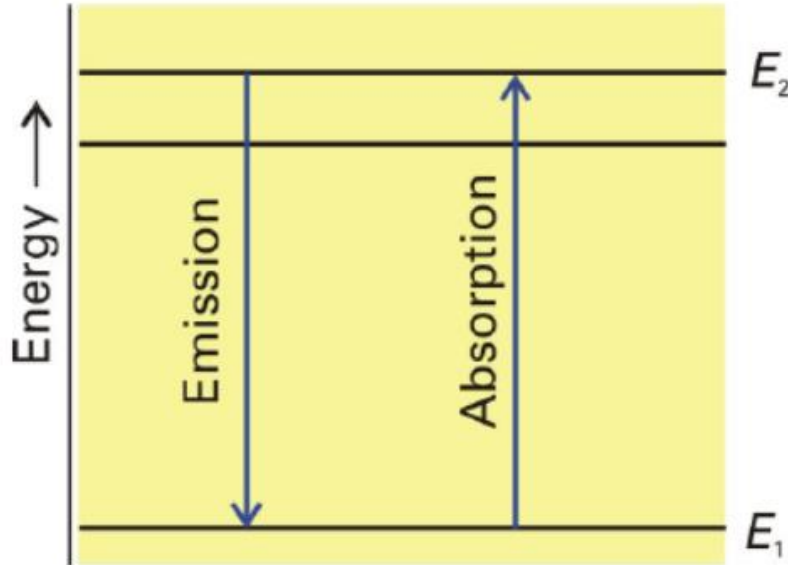
- *Electromagnetic radiation (EMR) includes, in addition to what we commonly refer to as “light”, radiation of longer and shorter wavelengths.*
- *EMR contains both an electric and magnetic component – plane polarized radiation.*
- *Electric field of strength of E .*
- *Magnetic field of strength of H*

$$E_y = A \sin(2\pi\nu t - kx)$$

$$H_z = A \sin(2\pi\nu t - kx)$$

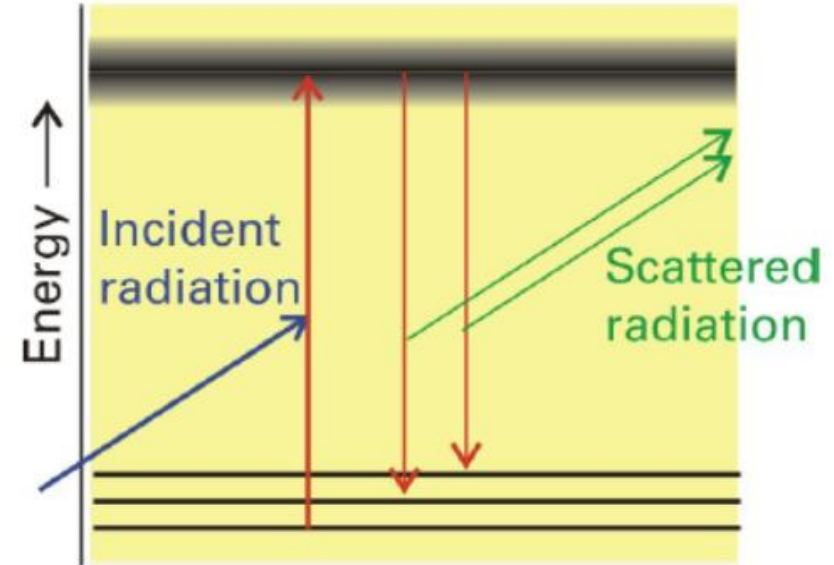


Types of interaction with electromagnetic radiations



Absorption or emission of the radiation by matter:

Absorption or emission spectroscopies
(pure rotational, vibrational, electronic, NMR, EPR spectra, etc)



Scattering of the radiation by matter:

Raman spectroscopies:
Raman rotational spectra,
Raman vibrational spectra, etc

Interaction of EMR and molecule

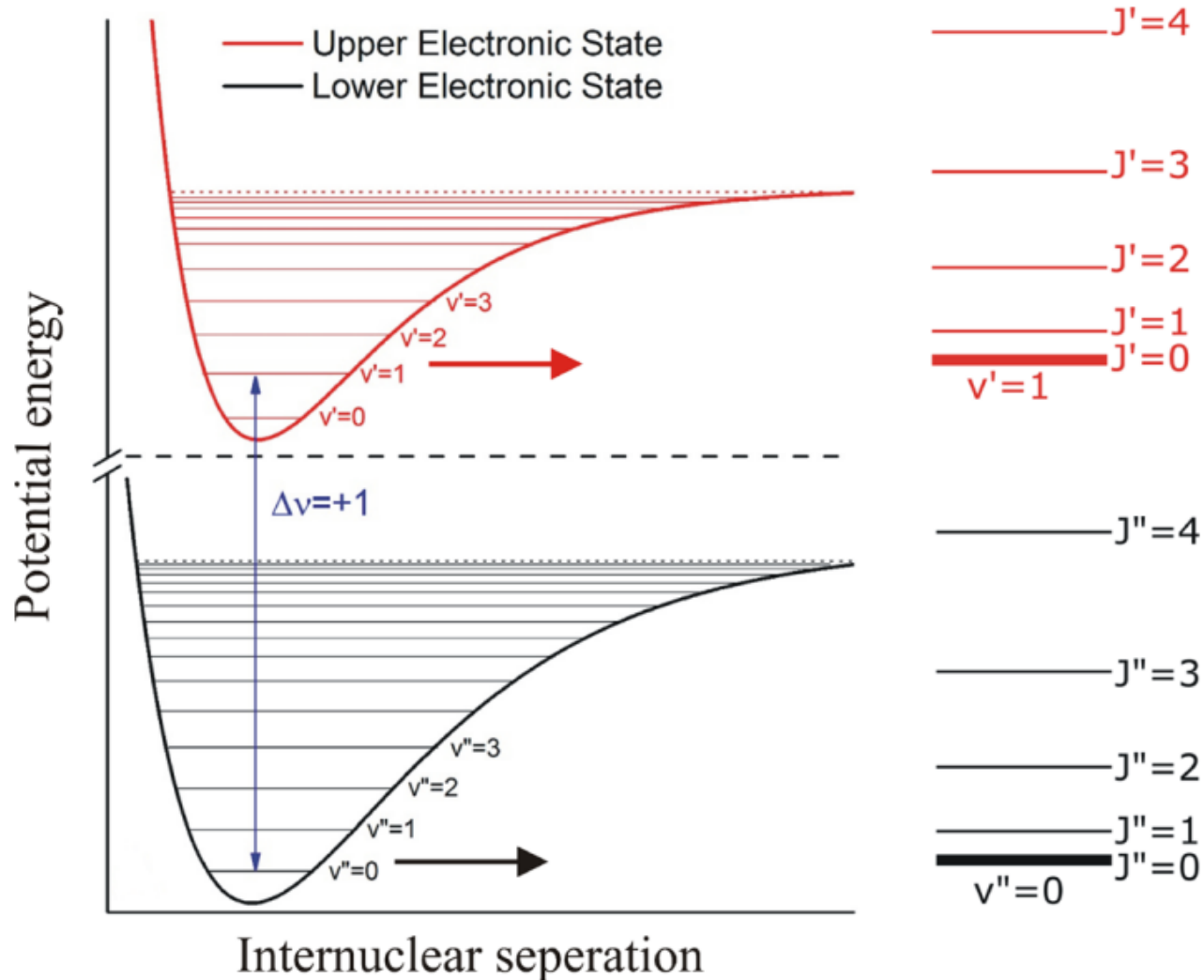
- *Electric/magnetic component of EMR may interact with the molecule.*
- *How?*
- *OSCILLATING ELECTRIC OR MAGNETIC DIPOLE GENERATES EMR AND ALSO ABSORBS EMR.*
- *Electric component may interact with the electric dipole moment of the molecule: e.g., Rotational-, vibrational-, and electronic- spectroscopies.*
- *Magnetic component may interact with the magnetic dipole moment of the molecule: e.g., NMR-, EPR-spectroscopies.*

Energy states of a molecule



- Within the limit of Born-Oppenheimer approximation, we can construct potential energy surfaces for different molecular electronic states.
- For polyatomic molecules, this can form a complex n-dimensional surface due to multiple interatomic distances.
- But, for the simplest case of a diatomic molecule, the electronic potential energy surface only depends on a single internuclear separation.
- The energy level structure of a simple molecular system such as a diatomic molecule can be described using the anharmonic oscillator—non-rigid rotor model framework with the first correction terms from the harmonic oscillator rigid-rotor approximation.

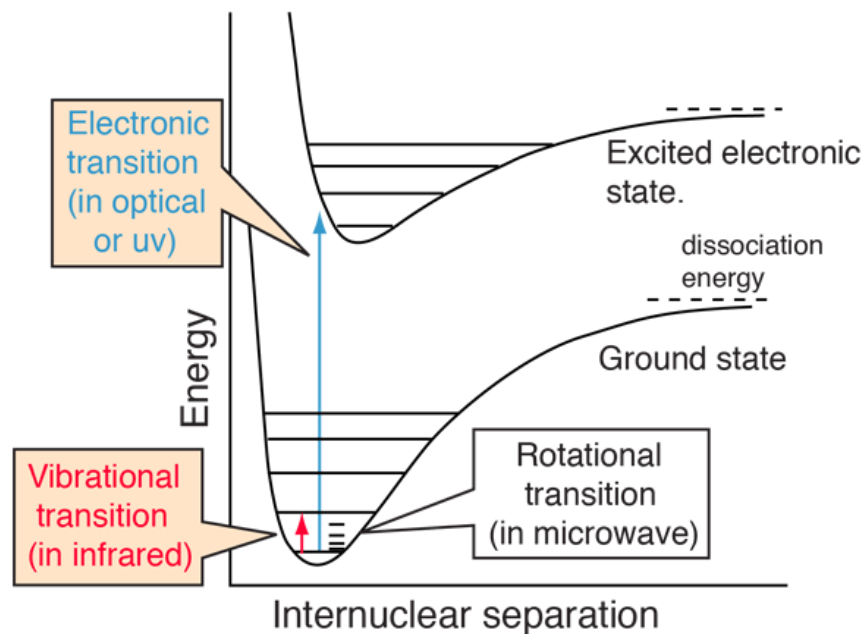
Energy states of a diatomic molecule



Energy states and transitions of a molecule



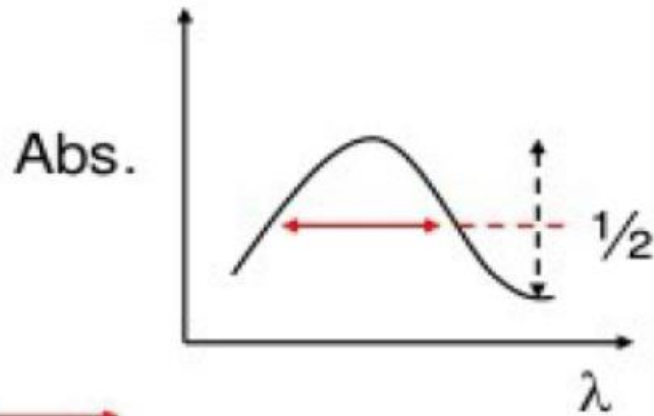
- Similarly to atomic transitions, each molecular species absorbs or emits light at a set of discrete and characteristic frequencies determined by the energy levels and their populations.
- In general, each molecular species has many more transitions than the corresponding atoms composing the molecule, due to the inclusion of vibrational and rotational energy levels.
- It is worth noting that for molecules with large moments of inertia, the rotational levels become more closely spaced and may be difficult to resolve using standard absorption and emission techniques.



Atomic versus Molecular spectra

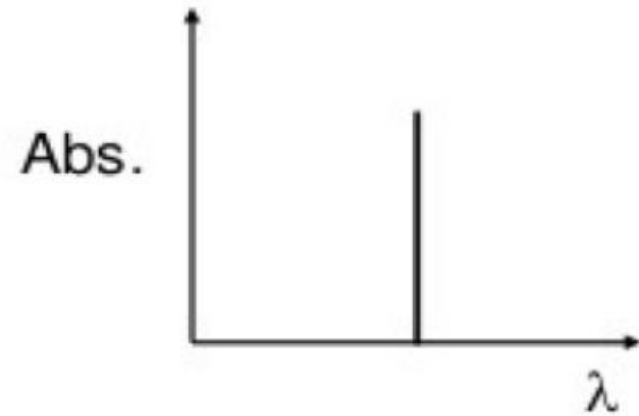


molecular
absorption band



band width ≥ 25 nm

atomic
absorption band



0.003 nm

- Atomic spectra contain sharp lines because of pure electronic transitions (only electronic spectra possible for atoms)
- Molecular spectral peaks have some band-width – mixing of various degrees of freedom

Some important terminologies

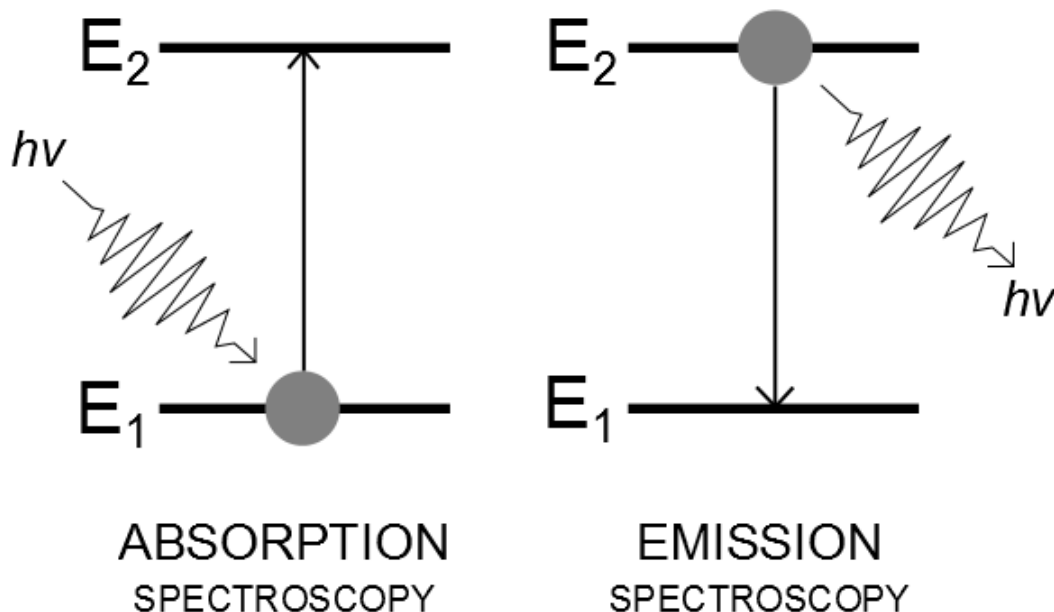


- **Signal to noise ratio** – Random fluctuations in the background caused by electronic signals in the recorded spectrum are termed as noise. A real spectral peak should be sufficiently intense so as to get distinguished from the noise
- **Resolution** – Smallest observable separation between bands – in terms of wavelength/frequency/wave-numbers
- **Fourier transform technique (FT)** – Used in various spectroscopies (e.g. FT-IR, FT-NMR, FT-Raman, etc. Provides simultaneous and almost instantaneous recording of the whole spectrum. Offers greater sensitivity and speed
- **Selection rules** – Gross and specific selection rules:
 - **Gross selection rules:** Features that molecule must possess so that the given spectroscopic technique can analyse the molecule
 - **Specific selection rules:** Which transitions are allowed/forbidden?

Electronic Spectroscopy:

UV-Visible absorption spectroscopy and Emission spectroscopy

Absorption can only occur when the energy of the radiation (calculated from the frequency or wavelength) matches the energy gap.



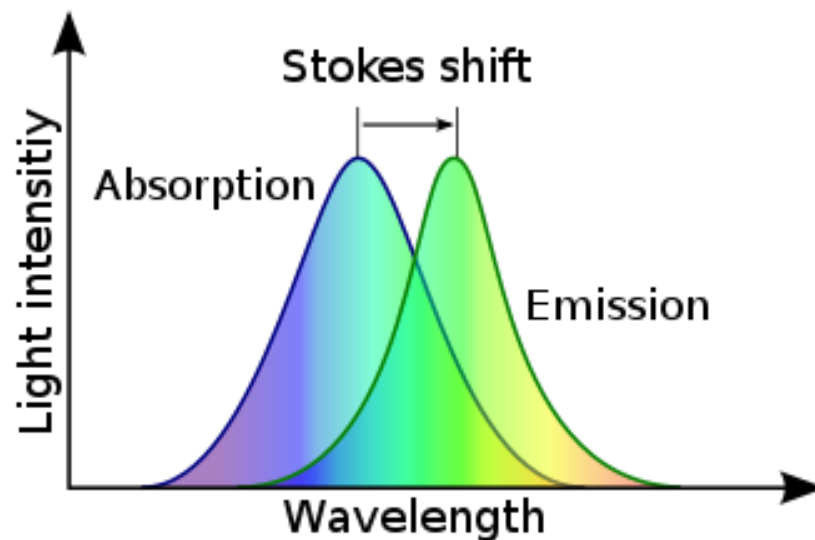
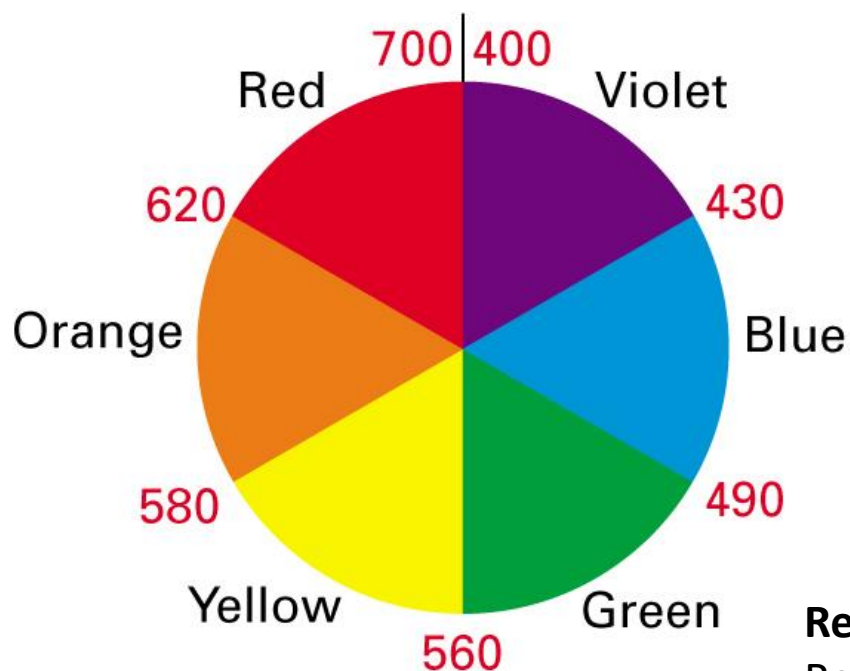
Absorbance: *Provides information about the ground state of the molecule.*

Emission: *Provides information about the excited state of the molecule.*

Absorption and emission between two electronic levels
(a simplified diagram)

Electronic Transitions and Spectra

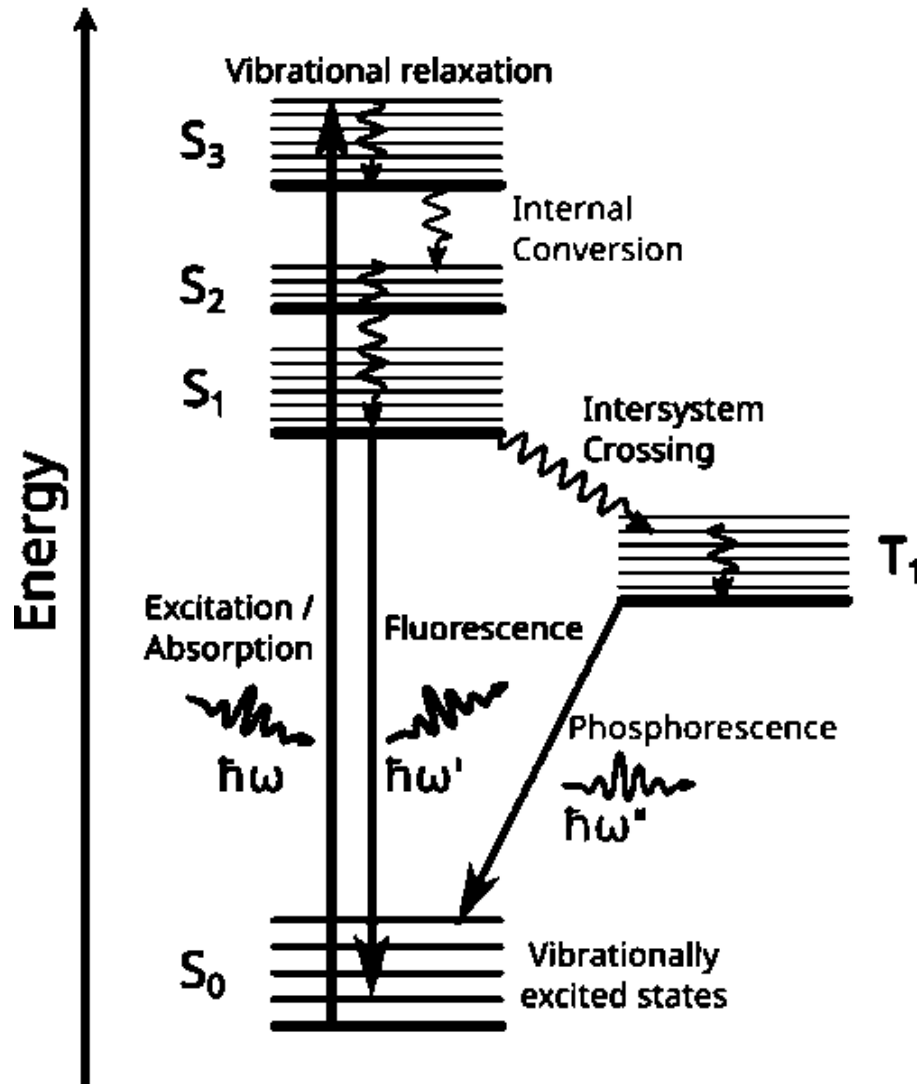
- Occur in the visible (400 to 700 nm) and ultraviolet (200 to 400 nm) regions of spectrum.*
- Responsible for the color of many objects, many photochemical reactions.*



Recollect your knowledge:
Red shift and blue shift

Spectrum: singular
Spectra: plural

Jablonski diagram



Non-Radiative processes:

- *Disordered thermal motion of surroundings of the molecule*
- *Internal Conversion.*
- *Intersystem crossing*

Radiative processes:

- *Fluorescence*
- *Phosphorescence*

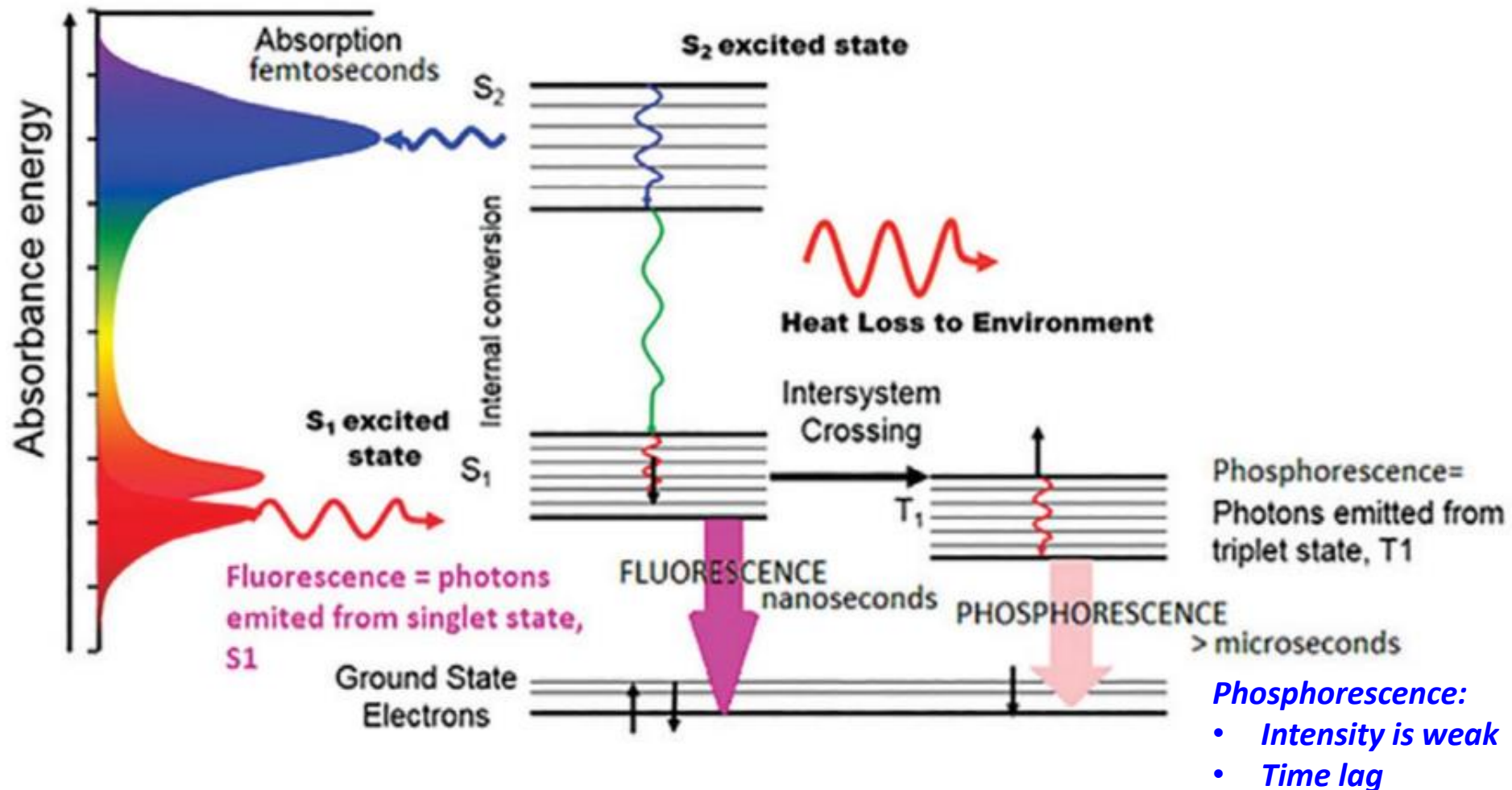
Jablonski diagram

innovate

achieve

lead

Jablonski Diagram for Fluorescence and Phosphorescence



Time scales of various processes



Transition	Process	Rate constant	Timescale (s)
$S_0 \rightarrow S_1$ or S_n	Absorption (Excitation)		10^{-15}
$S_n \rightarrow S_1$	Internal conversion	k_{ic}	10^{-14} to 10^{-10}
$S_1 \rightarrow S_1$	Vibrational relaxation	k_{vr}	10^{-12} to 10^{-10}
$S_1 \rightarrow S_0$	Fluorescence	k_f	10^{-9} to 10^{-7}
$S_1 \rightarrow T_1$	Intersystem crossing	k_{isc}	10^{-10} to 10^{-8}
$T_1 \rightarrow S_0$	Phosphorescence	k_p	10^{-3} to 10^0

UV-visible absorption and emission spectra of some molecules

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